

Maths.

Q.1

(a) $16^{\frac{3}{4}} \times 16^{\frac{1}{4}}$

$$= 16^{\frac{3}{4} + (\frac{1}{4})}$$

$$= 16^{\frac{3+1}{4}}$$

$$= 16^{\frac{4}{4}}$$

$$= 16^1$$

$$= (4^2)^1$$

$$= 4$$

{ Answer }

(b) $\frac{(x^{-2}y^3)^{-2}}{(x^3y^{-1})^2}$

$$= \frac{x^{-2 \times -2} \cdot y^{3 \times -2}}{x^{3 \times 2} \cdot y^{-1 \times 2}}$$

$$= \frac{x^{+4} \cdot y^{-6}}{x^6 \cdot y^{-2}}$$

$$= \frac{x^{4-6} \cdot y^{-6+2}}{x^{-2} \cdot y^{-4}}$$

$$= x^{-2} \cdot y^{-4}$$

$$= \frac{1}{x^2 y^4}$$

$$= \frac{1}{x^2 y^4}$$

Q.2

$$\textcircled{a} \log \left[\frac{P^3 \sqrt{q}}{r^4} \right]$$

$$\log \left(\frac{P^3 \times (q)^{\frac{1}{2}}}{r^4} \right)$$

$$\log P^3 + \log a^{\frac{1}{2}} - \log r^4$$

$$3 \log P + \frac{1}{2} \log q - 4 \log r$$

$$\textcircled{b} \log 0.9$$

$$\log \left(\frac{9}{10} \right)$$

$$\log \frac{3^2}{10}$$

$$\log 3^2 - \log 10$$

$$2 \log 3 - \log 10$$

$$2(0.4771) - 1$$

$$0.9542 - 1$$

$$= -0.0458$$

Q.3

$$\text{first polynomial} :- x^2 + 2x - 24$$

$$\text{HCF} :- (x - 4)$$

$$\text{LCM} :- x^3 + x^2 - 26x + 24$$

formula :-

$$P \times Q = \text{HCF} \times \text{LCM}$$

$$Q = \frac{\text{HCF} \times \text{LCM}}{P}$$

$$Q = \frac{(x-4)(x^3+x^2-26x+24)}{x^2+2x-24}$$

$$= \frac{(x-4)(x^3+x^2-26x+24)}{x^2+6x-4x-24}$$

$$= \frac{(x-4)(x^3+x^2-26x+24)}{x(x+6)-4(x+6)}$$

$$= \frac{(x-4)(x^3+x^2-26x+24)}{(x-4)(x+6)}$$

$$= \frac{x^3+x^2-26x+24}{(x+6)}$$

divide :-

	$x^2 - 5x + 4$	
$x+6$	$x^3 + x^2 - 26x + 24$	$x+6$
x	$\oplus x^3 + 6x^2$	x^2
	$\ominus$	$\frac{x^3+6x^2}{x^3+6x^2}$
	$-5x^2 - 26x$	5
	$\oplus 5x^2 + 30x$	$x+6$
	$\ominus$	$5x$
	$+4x+24$	$\frac{5x}{5x^2+30}$
	$\oplus 4x + 24$	
	$\ominus$	
	x	

$$Q: x^2 - 5x + 4$$

$$Q: 3(iv)$$

$$\frac{1}{x^2 - 1} + \frac{1}{x^2 - 3x + 2} + \frac{1}{x^2 + 3x + 2}$$

$$\frac{1}{(x)^2 - (1)^2} + \frac{1}{x^2 - 2x + 1} + \frac{1}{x^2 + 2x + 1}$$

$$\frac{1}{(x-1)(x+1)} + \frac{1}{x(x-2) - 1(x-2)} + \frac{1}{x(x+2) + 1(x+2)}$$

$$\frac{1}{(x-1)(x+1)} + \frac{1}{(x-1)(x-2)} + \frac{1}{(x+1)(x+2)}$$

$$= \frac{(x-2)(x+2)}{(x-1)(x+1)(x-2)(x+2)}$$

$$= \frac{(x+2)(x-2)}{(x-1)(x+1)} + \frac{(x+1)(x+2)}{(x-1)(x-2)} + \frac{(x-1)(x-2)}{(x+1)(x+2)}$$

$$= \frac{(x+2)(x-2)}{(x+1)(x+2)(x-1)(x-2)} + \frac{(x+1)(x+2)}{(x+1)(x+2)(x-1)(x-2)} + \frac{(x-1)(x-2)}{(x+1)(x+2)(x-1)(x-2)}$$

$$= \frac{(x^2 - 4)}{(x+1)(x+2)(x-1)(x-2)} + \frac{(x^2 + 3x + 2)}{(x+1)(x+2)(x-1)(x-2)} + \frac{(x^2 - 3x + 2)}{(x+1)(x+2)(x-1)(x-2)}$$

$$= \frac{n^4 - 4n^3 + 3n^2 + 2n^2 - 3n + 2}{(n-1)(n+1)(n-2)(n+2)}$$

$$= \frac{n^4 + n^2 + n^2 - 4 + 2 + 2 + 3n - 3n}{(n-1)(n+1)(n-2)(n+2)}$$

$$= \frac{3n^2 - 4}{(n^2 - 1)(n^2 - 4)}$$

Q.2(v)

(a) $|2n+1| = 13$

$$2n+1 = 13 \quad , \quad 2n+1 = -13$$

$$2n = 13-1 \quad , \quad 2n = -13-1$$

$$2n = 12 \quad , \quad 2n = -14$$

$$n = \frac{12}{2} = 6 \quad , \quad n = \frac{-14}{2} = -7$$

$$n = 6, -7$$

$$\{6, -7\}$$

(b) $-7 < 3n+2 \leq 14$

$$-7-2 < 3n+2-2 \leq 14-2$$

$$-9 < 3n \leq 12$$

$$\frac{-9}{3} < n \leq \frac{12}{3}$$

$$-3 < n \leq 4$$

$$-3 < n \leq 4$$

$$\{n/n \in \mathbb{R}, -3 < n \leq 4\}$$

$$g: 2(v_i)$$

$$A(5, 7)$$

$$B(2, 3)$$

$$C(10, -3)$$

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(2-5)^2 + (3-7)^2}$$

$$= \sqrt{(-3)^2 + (-4)^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

$$= 5$$

$$d_{BC} = \sqrt{(10-2)^2 + (-3-3)^2}$$

$$= \sqrt{(8)^2 + (-6)^2}$$

$$= \sqrt{64+36}$$

$$= \sqrt{100}$$

$$= 10$$

$$d_{AC} = \sqrt{(5-10)^2 + (7-(-3))^2}$$

$$= \sqrt{(-5)^2 + (10)^2}$$

$$= \sqrt{25+100}$$

$$= \sqrt{125}$$

$$= \sqrt{125}$$

checking if it is right angled triangle with the help of pythagoras theorem.

$$AB^2 + BC^2 = AC^2$$

$$(5)^2 + (10)^2 = (\sqrt{125})^2$$

$$25 + 100 = 125$$

$$125 = 125$$

AB and BC meet at B, the right angle lies at $\therefore \angle B = 90^\circ$

Q: 3

(i)

$$\frac{n^2 - 25}{n^2 + 8n + 15} \times \frac{n^2 + 3n}{n^2 - 3n - 10}$$

$$\frac{(n)^2 - (5)^2}{n^2 + 3n + 5n + 15} \times \frac{n(n+3)}{n^2 + 2n - 5n - 10}$$

$$\frac{(n-5)(n+5)}{n(n+3) + 5(n+3)} \times \frac{n(n+3)}{(n+2) - 5(n+2)}$$

$$\frac{(n-5)(n+5)}{(n+5)(n+3)} \times \frac{n(n+3)}{(n-5)(n+2)}$$

$$\text{Answer: } \left\{ \frac{n}{n+2} \right\}$$

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$$3n^2 - 2n + 1$$

$$\begin{array}{r}
 3n^2 \overline{) 9n^4 - 12n^3 + 10n^2 - 4n + 1} \\
 \underline{9n^4} \\
 0n^3 + 2n \\
 \underline{2n} \\
 0n^3 + 4n + 1 \\
 \underline{4n} \\
 0n^3 + 0n + 1 \\
 \underline{0n^3 + 0n + 0} \\
 1
 \end{array}$$

Q: 3(ii) $P(1, 2), Q(4, 6), R(7, 10)$

a) d_{PQ}

$$\begin{aligned}
 d_{PQ} &= \sqrt{(4-1)^2 + (6-2)^2} \\
 &= \sqrt{3^2 + 4^2} \\
 &= \sqrt{9 + 16} \\
 &= \sqrt{25} \\
 &= 5
 \end{aligned}$$

$$\begin{aligned}
 d_{QR} &= \sqrt{(7-4)^2 + (10-6)^2} \\
 &= \sqrt{3^2 + 4^2} \\
 &= \sqrt{9 + 16} \\
 &= \sqrt{25} \\
 &= 5
 \end{aligned}$$

$$\begin{aligned}
 d_{PR} &= \sqrt{(7-1)^2 + (10-2)^2} \\
 &= \sqrt{(6)^2 + (8)^2} \\
 &= \sqrt{36 + 64} \\
 &= \sqrt{100} \\
 &= 10
 \end{aligned}$$

(b) Collinear or not :-

$$\begin{aligned}
 AB + BC &= AC \\
 5 + 5 &= 10 \\
 10 &= 10
 \end{aligned}$$

(c) midpoint of PR :-
 $P(1, 2)$, $R(7, 10)$

midpoint {formula} :-

$$\frac{x_2 + x_1}{2} , \frac{y_2 + y_1}{2}$$

$$= \frac{7+1}{2} , \frac{10+2}{2}$$

$$= \frac{8}{2} , \frac{12}{2}$$

$$\therefore (4, 6)$$